Assignment 7D. Finite Element Method Marks 20

Posted on 28.10.2026 @ 2:30 pm and due on 28.10.2026 @ 5:40 pm

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1. A copper rod of length 2 meter is clamped rigidly at its two end kept at constant temperatures $y(x = 0) = 40^o$ and $y(x = 2) = 100^o$ Celsius. A current is flowing through the rod that generates a constant heat $f(x) = -50$ unit. The governing equation is

$$\frac{d^2y(x)}{dx^2} = f(x)$$

   Show, by plotting, the steady state temperature profile $y(x)$ using FEM. **[8]**

2. Solve 1+1 dimension heat equation

$$\frac{\partial^2 u(x,t)}{\partial x^2} = \frac{\partial u(x,t)}{\partial t}$$

   on a metal rod of length 2 units, with the initial conditions

$$\begin{aligned} u(0,t) = 0^o\text{C} = u(2,t) \qquad & \text{for} \quad 0 \le t \le 4 \\ u(x,0) = 20\,|\sin(\pi x)|^o\text{C} \qquad & \text{for} \quad 0 \le x \le 2 \end{aligned}$$

   Use explicit scheme taking number of position grid $Nx = 20$ and time grid $Nt = 2000$. Show the temperature profile across the length of the rod at time steps 0, 10, 20, 50, 100, 200, 500 and 1000 in a plot. **[12]**